

Aim

To write a C program to simulate a DFA and an NFA that accept binary strings ending with the substring "01".

Algorithm

1. **Start the program and read a binary string as input.**
2. **DFA Simulation:**
 - Use a variable to track the current state.
 - Update state based on input character.
 - Final state is accepting if it indicates the string ends with '01'.
3. **NFA Simulation:**
 - Check only the last two characters of the string.
 - If they are '0' and '1', the string is accepted.
4. **Print acceptance or rejection messages for both simulations.**

C Program Code

```
#include <stdio.h>
#include <string.h>

// DFA: accepts if string ends with 01
int dfa_accepts(char str[]) {
    int state = 0;
    int i = 0;

    while (str[i] != '\0') {
        char ch = str[i];
        if (state == 0) {
            if (ch == '0') state = 1;
            else state = 0;
        } else if (state == 1) {
            if (ch == '1') state = 2;
            else state = 1;
        } else if (state == 2) {
            if (ch == '0') state = 1;
            else state = 0;
        }
    }
}
```

```

        i++;
    }

    return (state == 2);
}

// NFA: checks if last two characters are 0 and 1
int nfa_accepts(char str[]) {
    int len = strlen(str);
    if (len < 2) return 0;
    return (str[len - 2] == '0' && str[len - 1] == '1');
}

int main() {
    char input[100];
    printf("Enter a binary string: ");
    scanf("%s", input);

    printf("\nUsing DFA: ");
    if (dfa_accepts(input))
        printf("Accepted (ends with 01)\n");
    else
        printf("Rejected\n");

    printf("Using NFA simulation: ");
    if (nfa_accepts(input))
        printf("Accepted (ends with 01)\n");
    else
        printf("Rejected\n");

    return 0;
}

```

Sample Output

Enter a binary string: 1101

Using DFA: Accepted (ends with 01)

Using NFA simulation: Accepted (ends with 01)